GSE 518
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Problem Set 2

You must turn in problems: 1(a); 2 or 3; 4 and 5. However, you are strongly encouraged to work on all the problems before the midterm, since the material they cover will be part of the midterm. Please let me know if you have any questions.

1. For the following random experiments find the probability p of the event A.

   (a) Tossing a die, where the event A occurs if a five or six appears.

   (b) The choice of a point from the interior of a square with opposite vertices $(-1,-1)$ and $(1,1)$, where the event A occurs if the sum of the coordinates of the point is less than $1/2$.

2. A fair coin is tossed until a head appears.

   (a) What is the sample space?

   (b) What is the probability that a head will first appear on the third toss?

   (c) What is the probability that *exactly* x tosses will be required to produce the first head?

3. Two fair dice are tossed. Define the following events: $A = \{\text{odd face with first die}\}$, $B = \{\text{odd face with second die}\}$ and $C = \{\text{odd sum}\}$. (Note that "odd sum" is the same as saying "one face is odd, one face is even").

   (a) Show that the events A, B and C are pairwise independent.

   (b) Show that the events A, B and C are not mutually independent.

   (c) Now define the following events: $D = \{\text{doubles appear}\}$, $E = \{\text{the sum is between 7 and 10}\}$ and $F = \{\text{the sum is 2 or 7 or 8}\}$. Show that $P(D \cap E \cap F) = P(D)\,P(E)\,P(F)$. Are events D, E and F mutually independent?

4. Box I contains 3 red balls and 7 blue balls. Box II contains 6 red balls and 4 blue balls. A box is selected at random (each with equal probability) and then 1 ball is drawn from this box.

   (a) What is the probability that the ball drawn is red?

   (b) Given that this ball is red, what is the conditional probability that it came from box II?

5. Your friend chose at random to visit one of three cities: Seattle, Las Vegas and New York (each with equal probability). The probability of rain in Seattle was $1/3$; in Las Vegas was $1/10$ and in New York it was $1/5$. She came back with mud in her shoes and with a wet umbrella. Given this new information, what is the probability that she visited each of the cities?

6. *(Just for fun problem, i.e., you do not need to turn it in).* If two fair dice are tossed, what is the smallest number of throws for which the probability of getting at least one double six exceeds 0.5? Pascal and Méré corresponded about this problem in 1654. (*Hint:* For this problem is easier to find first the probability that *no* double six appears in n throws.)